An Evaluation Artefact in Presence–Background Archaeological Modelling: Evidence from East Java and a Corrected Comparison Protocol

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Abstract

Presence–background (presence-only) archaeological predictive models are usually evaluated by cross-validation against the same background points used to fit them. We show that this can manufacture an apparent performance gain where none exists. We built a settlement suitability model for East Java (378 presences; XGBoost, RandomForest and a Maximum Entropy model under fixed spatial-block cross-validation) and originally reported a monotone AUC ladder, which we attributed to increasingly realistic pseudo-absence (background) design. A benchmark requested by a reviewer refuted that attribution: held to a common evaluation background, redesigning the background contributes approximately nothing, whereas adding a single hydrological feature contributes +0.042 **AUC**. The reported gain is an artefact of evaluation — each design was scored against the negatives it had selected for itself. Synthetic worlds of known ground truth show why: the selection criterion computed on a design's own background has *no interior optimum*, and reported and true performance move in opposite directions on the real data. We withdraw the taphonomic interpretation built on the ladder, correct two defects in our own

diagnostics (an incomplete volcano inventory and a non-reproducing correlation), and propose a corrected protocol: hold the evaluation background fixed, declare the availability domain of every metric, and seed-ensemble priority maps. The contribution is the diagnosis and the protocol, not a new site-prediction claim.

Keywords: presence-background modelling; pseudo-absence design; evaluation artefact; target-group background; spatial cross-validation; survey bias; East Java

1 Introduction

Predictive settlement modelling in Java is difficult because the observed archaeological record is shaped as much by preservation and discovery as by past settlement behaviour. Known site maps are incomplete and strongly conditioned by survey history; active volcanism and tephra deposition can bury older surfaces; and from a computational standpoint the combination produces label noise in pseudo-absence design and inflates validation when spatial dependence is mishandled. This paper addresses a narrower question than it first appears to: *how do you know that redesigning the background points improved a presence–background model?* The answer we offer is uncomfortable, because it is the answer we had to give about our own model.

1.1 What presence–background models are, and why their evaluation is delicate

Presence–background (presence-only) models learn a suitability surface from known presences and a set of background points that stand in for the available environment. They are the workhorse of ecological niche modelling (ENM) and species distribution modelling (SDM) (Elith et al. 2006; Phillips, Anderson & Schapire 2006) and of much archaeological predictive modelling (Kvamme 2006; Verhagen & Whitley 2012). They are acutely sensitive to sample-selection bias, especially when the background does not reflect the same observation process as the presences (Araújo & Guisan 2006; Lobo, Jiménez-Valverde & Hortal 2010; Phillips et al. 2009; Wisz & Guisan 2009). The standard correction for

survey-biased data is target-group background (TGB): draw background points from the area a survey could plausibly have reached, so that the model learns presence against a realistic "available" set rather than against the whole landscape (Phillips et al. 2009).

A model improved in this way is judged by cross-validation against held-out points, and the held-out *negatives* are drawn from the same background distribution the model was trained on. This is where the delicacy lies. If two background designs are each scored against their own background, the comparison is not identified: a design that makes its own background easier to separate from presences will look better, for a reason that has nothing to do with how well either model captures the phenomenon. Random train/test splits overestimate transfer when observations are spatially clustered, because held-out points sit close to training points (Hijmans 2012; Roberts et al. 2017); spatially structured cross-validation is the recommended corrective (Valavi et al. 2019). But spatial blocking fixes where the folds are; it does not, by itself, fix *which* background the held-out negatives come from. That second choice is the subject of this paper.

1.2 Four reasons a site may be absent, and which one we model

Following a reviewer's taxonomy, a site may be absent from the record for four distinct reasons, and conflating them is the central interpretive hazard of this literature:

1. **Environmentally unsuitable** — the location was never attractive for settlement.
2. **Not surveyed** — a site could be there, but no one has looked there.
3. **Buried** — a site was there and is now below the detectable surface.
4. **Destroyed** — a site was there and has since been erased.

This paper models suitability only (mechanism 1). The other three are the interpretive frame, not the output of the model. A low suitability score does not distinguish "never settled" from "buried" or "destroyed"; a high suitability score does not assert that a site exists. We state this restraint up front because the earlier version of this manuscript overreached it, and the retraction of that overreach is reported in Section 4.

1.3 Contribution and relation to prior work

A reviewer of the previous version asked us to benchmark against Maximum Entropy (Elith et al. 2006), and that request turned out to be the decisive analysis. The benchmark refuted our own central claim (Section 3.2): the AUC ladder we attributed to background realism is an evaluation artefact. We have therefore not revised the paper's argument; we have replaced it. The same empirical material now supports a narrower and, we believe, more useful proposition: *background designs in presence–background modelling cannot be compared unless the evaluation background is held fixed, and a selection criterion computed on a design's own background has no interior optimum.*

Our contribution sits *inside* the ENM literature rather than beside it, and we state its boundary precisely, because a reviewer of the previous version observed — correctly — that this family of problems is well established in adjacent fields.

It is established. Lobo, Jiménez-Valverde & Real (2008) showed that AUC is not comparable between models whose background differs, because the metric is computed over whatever availability domain the modeller supplied; Barve et al. (2011) formalised that domain as the accessible area M and showed that its delineation conditions both the fitted model and its evaluation; and Hijmans (2012) demonstrated that the spatial arrangement of background relative to presences inflates cross-validated AUC, proposing a null-model calibration for it. Model-selection criteria computed within a single background are likewise known to be unreliable guides to transferability (Radosavljevic & Anderson 2014; Warren & Seifert 2011), and discrimination metrics in general are known to answer a narrower question than applications ask of them (Guillera-Arroita et al. 2015; Lobo, Jiménez-Valverde & Hortal 2010). That the target-group correction is standard (Phillips et al. 2009) and that pseudo-absence choice moves reported performance (Barbet-Massin et al. 2012; Senay, Worner & Ikeda 2013) are both settled points. The transfer of species-distribution modelling into archaeology is equally established — from eco-cultural niche modelling (Banks et al. 2006), through MaxEnt-based prediction of archaeological site distributions (Howey, Palace & McMichael 2016; Noviello et al. 2018; Yaworsky et al. 2020), to the standard methodological reference (Franklin 2009) — and

recent machine-learning work in archaeology reports strong gains from integrating spatial context and terrain logic while flagging data-quality and validation constraints (Castiello & Tonini 2021; Comer et al. 2023; Wang, Shi & Oguchi 2023).

Three things here are ours, and none of them is "background design matters". First, we execute on one dataset the comparison this literature implies but rarely runs end to end: the same designs, the same folds, scored both on their own background and on a common one, so the artefact is *measured* rather than argued (Section 3.2). Second, we characterise the criterion a modeller actually optimises — AUC on the design's own background — and show it has *no interior optimum*, so it cannot tell them when to stop; we are not aware of a comparable dose-response measurement of reported against true performance along a background-design dial (Section 3.3). Third, we give the effect a size in an archaeological setting: the inflation is of the same order as the gain that we, and others, report as a result — which is why this is a substantive hazard and not only a metrological caveat.

The honest shape of this revision is worth stating. A title can be softened to match weaker evidence, but when the evidence supports a different proposition, softening is not enough; the proposition has to be replaced. Retitling from a bias-correction claim to an evaluation-artefact finding is that kind of replacement, and we believe it is what the reviewer's request required.

1.4 Study area

The study area is province-scale East Java (approximate bounds 111–115°E, 9–6.5°S), processed in EPSG:32749. East Java's archaeological record is dominated by the Hindu–Buddhist period (roughly the 8th–15th centuries CE), including the Singhasari and Majapahit kingdoms, whose temples and inscriptions concentrate in the Brantas river valley around Malang–Mojokerto–Blitar (e.g. Candi Penataran, Singhasari, Jago and Kidal). The known record is therefore both chronologically and spatially selective: it samples the well-surveyed lowlands more densely than either the volcanic highlands (Kelud, Semeru, Bromo–Tengger) that surround them or the karstic southern mountains, and discovery

history is strongly conditioned by modern road access. The 378 usable presences in this study are the covariate-valid subset of that record; Figure 1 shows how unevenly they cover the province.

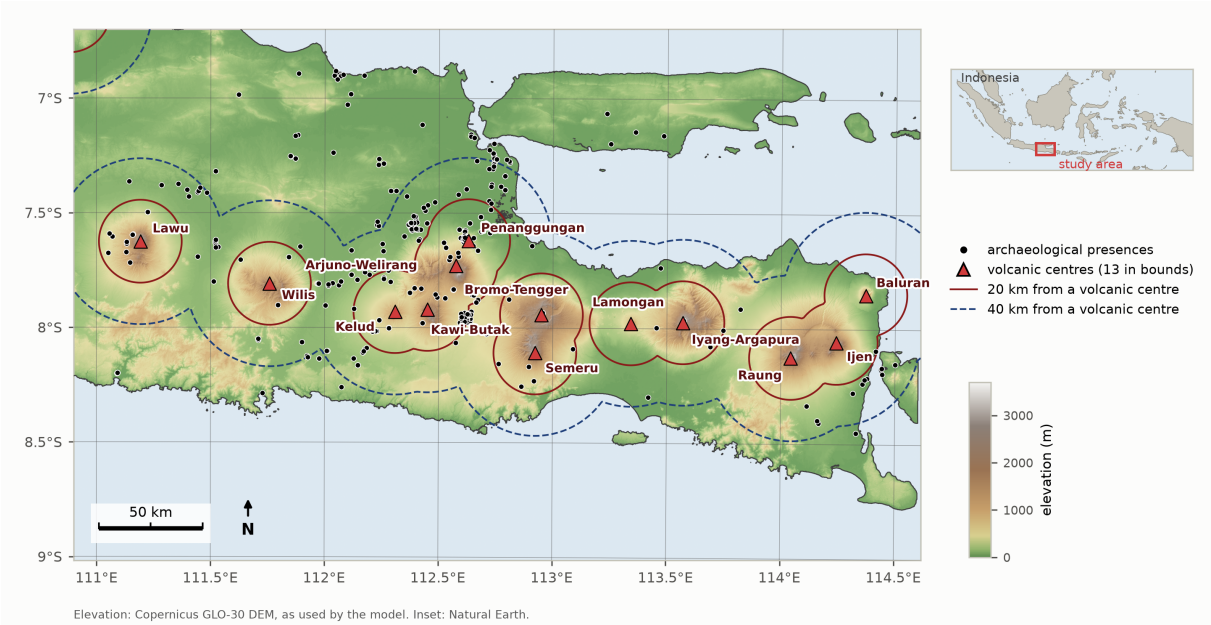


Figure 1: Study area in East Java, Indonesia, showing the presences used for modelling and the 13 canonical volcanic centres inside the paper's stated bounds (111–115°E), used as post-hoc diagnostics only.

2 Materials and Methods

2.1 Data, covariates, and the principle of role separation

Usable presences after covariate filtering were 378 (111–115°E); the soil-coverage subset used once in E009 was 259. Every variable plays one analytical role, and the roles must not be mixed, because mixing them makes a suitability score uninterpretable: a low score could mean the place was unsuitable, or unsurveyed, or not preserved, and the model cannot say which. We therefore keep only the suitability block as predictors, express accessibility through the background design (Section 2.4), and treat preservation and volcanic proximity as diagnostics applied to the output. Table 1 assigns every variable a single role.

This separation is not a technicality. After the benchmark reported in Section 3, it is

Variable	Source	Analytical role	How it enters
elevation	Copernicus GLO-30 DEM, 30 m, decimated ×10 (∼300 m)	Suitability (terrain)	training feature
slope	derived from DEM	Suitability (terrain cost)	training feature
TWI	topographic wetness index, derived from DEM	Suitability (water availability)	training feature
TRI	terrain ruggedness index, derived from DEM	Suitability (terrain cost)	training feature
aspect	derived from DEM	Suitability (insolation/orientation)	training feature
river_dist	OSM/HydroSHEDS, Euclidean distance	Suitability (water access); added at E008	training feature
clay, silt	SoilGrids topsoil	Preservation / substrate	features in **E009 only**; dropped (lowered AUC)
road_dist	OSM roads (expanded class set from E012)	Accessibility / survey effort	**never a training feature**
volcano_dist	canonical inventory, 30 centres (13 in bounds)	Taphonomy / burial — diagnostic only	**never a training feature**

Table 1: Covariates and their analytical roles. Roles are exclusive by construction: a suitability variable answers *would people have settled here?*; an accessibility variable answers *would we have found it?*; a preservation variable answers *would it still be there to find?*. TWI (topographic wetness index) quantifies the tendency of a cell to accumulate water; TRI (terrain ruggedness index) quantifies local elevation difference as a proxy for terrain cost.

the paper's actual subject, because the variable that defines the background (accessibility, via `road_dist`) is the variable that turned out to drive the reported number.

2.2 The modelling pipeline

We model binary presence/background suitability. The pseudo-absence (background) ratio is fixed at 1:5 throughout. Three learners are compared: XGBoost (Chen & Guestrin 2016) (primary), RandomForest (Breiman 2001), and a Maximum Entropy model (Elith et al. 2006; Phillips, Anderson & Schapire 2006) added at a reviewer's request. The Maximum Entropy model is the `elapid` implementation of maxnet, with linear, hinge and product features, regularisation multiplier $\beta = 1.5$ and a cloglog transform; β is swept from 0.5 to 4.0 in Section 3.2 to confirm the result is not an artefact of that setting. Validation uses 5-fold deterministic spatial block cross-validation at a baseline block size of 0.45° (∼50 km). Primary metrics are AUC and the True Skill Statistic (TSS), the latter preferred to kappa because it does not vary with prevalence in the way kappa does (Allouche, Tsoar & Kadmon 2006). Both are discrimination metrics computed against whatever background is supplied to them, which is precisely the property this paper is about (Guillera-Arroita et al. 2015; Lobo, Jiménez-Valverde & Real 2008). Our minimum viable result (MVR) criterion is spatial AUC > 0.75 under block cross-validation.

2.3 The experiment progression, as the object under examination

Seven iterative experiments (E007–E013) built the model whose evaluation we now examine. They are no longer the argument of the paper; they are the object the argument is about, so we summarise them in one table and two paragraphs and move the per-iteration detail to the supplement. Table 2 states exactly which covariates each experiment used, read from the analysis scripts rather than from any earlier prose.

Exp.	elev	slope	twi	tri	aspect	river	road	Background design	Rep. spatial AUC
E007 v1	✓	✓	✓	✓	✓	–	–	uniform random, 1:5, 2 km buffer	0.659
E008 v2	✓	✓	✓	✓	✓	✓	–	uniform random, 1:5	0.695
E009 v3	✓	✓	✓	✓	✓	✓	–	uniform random, 1:5 (+clay/silt)	0.664 ↓
E010 v4	✓	✓	✓	✓	✓	✓	bg	TGB (major road)	0.711
E011 v5	✓	✓	✓	✓	✓	✓	bg	TGB, parameters swept	0.725
E012 v6	✓	✓	✓	✓	✓	✓	bg	TGB, expanded road classes	0.730
E013 v7	✓	✓	✓	✓	✓	✓	bg	hybrid = TGB pool + regional blend + hard negatives	0.768 best / 0.751 seed-avg

Table 2: Per-experiment covariate inclusion. ✓ = training feature; – = not used; *bg* = used to construct the background, never as a feature. Reproducibility: rasters in `data/processed/dem/`, presences in `data/processed/east_java_sites.geojson` (378 valid in bounds), deterministic 5-fold spatial blocking at 0.45°, pseudo-absence ratio 1:5 throughout. E010–E013 use `road_dist` to build the background while excluding it from the features.

Two paragraphs of interpretation. Feature-only expansion plateaued (E007–E009): adding river distance helped (+0.036) and adding soil covariates hurt (−0.031). Background redesign then delivered the monotone rise (E010–E013) that we originally reported as evidence that pseudo-absence realism dominates feature accumulation. The remainder of this paper asks whether that rise is real.

The hybrid design of E013 blends a TGB candidate pool with a regional quota and a fraction of "hard negatives" (cells environmentally dissimilar from presences, $2.0 \leq z \leq 5.0$), sweeping `region_blend` $\in \{0.0, 0.3, 0.5, 0.7\}$ and `hard_frac` $\in \{0.0, 0.15, 0.30\}$. Note that the realised proportion of environmentally dissimilar pseudo-absences in the best configuration was 0.62, above the 0.30 target: the TGB pool, constrained to road-accessible cells, is intrinsically more dissimilar from site environments than unconstrained random sampling. This pool-composition effect bears on the absolute AUC values and is part of why a common evaluation background matters (Section 3.2).

2.4 The four roles of `road_dist`

`road_dist` is deliberately never a training feature, yet it does four jobs at once: it (i) defines the TGB acceptance probability, (ii) defines the hybrid background pool, (iii) defines the accessibility-proxy holdout split (E014, Section 2.5), and (iv) was a tautology proxy in Tests 1 and 3 of the earlier version's diagnostic suite. In that suite, Test 1 was the correlation between predicted suitability and three visibility proxies (volcano, road and nearest-site distance), failed only if $|\rho| > 0.5$; Test 3 checked that performance did not collapse in the least-surveyed road-distance quartile. The suite is no longer the spine of this paper; the corrected, seed-ensembled volcano-distance component of Test 1 is reported in Section 3.8. The submitted manuscript acknowledged this coupling once, in the limitations. Given the benchmark in Section 3, that placement was wrong: the variable that builds the background is the variable that drives the reported number, and it now appears here, where the covariates are introduced. This is also the honest answer to the question of what makes the approach specifically archaeological: the archaeological content is not in the label set, it is in the background design — a claim about where a survey could plausibly have detected a site had one been present.

2.5 The accessibility-proxy holdout (E014)

E014 trains on easy-access presences (`road_dist` ≤ 1 km) and tests on hard-access presences (> 1 km), using road accessibility as a proxy for discovery order: sites far from roads stand in for sites found late. We previously called this a "temporal split"; we now call it an *accessibility-proxy holdout*, because the split is defined by access, not by date, and the earlier label overstated what the data support. (The stored result file had itself been mislabelled with temporal headers; the underlying split is the accessibility split described here, AUC = 0.755 is unaffected, and the file now records which branch ran — see the Further Disclosures in the Response to Reviewers.)

2.6 The refutation design: E217–E224

The seven experiments that overturn the ladder were pre-registered (decision rules written before the runs) and are the paper's centrepiece.

One deviation from the published pipeline must be stated before the results, because it bears on how they should be read. E217–E224 draw background points from a decimated raster lattice (∼300 m spacing) rather than by the continuous-coordinate rejection sampling used in E007–E013. This was a deliberate choice: it gives every background design a single shared sampling frame, which is a precondition for the common-background comparison, and it keeps a factorial benchmark tractable. The cost is that absolute AUCs in E217–E224 are not directly comparable to the published E007–E013 values. The quantities we interpret are therefore *within-benchmark contrasts* — design against design, and own-background against common-background, under identical folds — and not the absolute levels. Because the decimation is ours and not the paper's, E218 reran the comparison at half that spacing (∼150 m, 5 seeds): the ladder persists on own backgrounds (random 0.711 → hybrid 0.757) and stays flat on the common background (random 0.705 → hybrid 0.704), so the conclusion does not depend on the lattice.

E217 – common-background benchmark (answers the Maximum Entropy request). MaxEnt, XGBoost and RandomForest are run under *identical* spatial-block folds and *identical* background designs as E007–E013, but each design is scored against a *common* evaluation background shared across designs, so that the comparison is identified. A second feature set (`terrain` alone, 5 vars vs `terrain_river`, 6 vars) lets us compare a feature effect against a background-design effect.

E218 – the evaluation artefact, isolated. Three training designs are each scored against four evaluation backgrounds (own + three foreign), exposing how much of each design's apparent lead comes from being judged at home.

E220 – does the selection criterion have an interior optimum? We sweep the hard-negative dial and ask whether the AUC-on-own-background criterion ever tells a

researcher to stop short of the dial's edge.

E221 – seed stability of the priority map. Across random seeds, how much of the top-decile priority map turns over, and how many seeds must be ensembled to stabilise it?

E222 – synthetic ground truth. Four synthetic worlds of known intensity surface let us compare *reported* AUC (own-background) against *true* recovery (common-background) where ground truth is not in question.

E223 – confidence intervals and regularisation. Bootstrap and equivalence CIs bound the ladder gain; a MaxEnt regularisation sweep ($\beta = 0.5$–4.0) tests whether the artefact is an overfit of an unregularised model.

E224 – a diagnostic we pre-registered and then withdrew (see Section 3.5). We proposed that the TGB null in E222 was structural, tested it, and the test failed; the null is reported as unexplained.

3 Results

3.1 The reported ladder

Reproducing the claim under examination: Table 3 and Figure 2 show the rise from E007 (0.659) to E013 (seed-averaged 0.751; best run 0.768; TSS 0.507 ± 0.167), with 20-seed stability in the range 0.729–0.774. Even the worst seed exceeded a spatial-interpolation null (DKNS, 0.646) by +0.083, and the model beat the three null models in Table 4. This is the performance we reported, and on its face it supports the conclusion we drew.

3.2 The ladder is an evaluation artefact

The rise is not real. On real data, redesigning the background while holding a common evaluation background fixed contributes −0.0142 **AUC** averaged over the three algo-

Exp.	Pseudo-absence strategy	XGB AUC	RF AUC	Best AUC	XGB TSS	RF TSS	Status
E007	Random	0.659 ± 0.077	0.656 ± 0.090	0.659	0.318 ± 0.126	0.314 ± 0.133	REVISIT
E008	Random + river feature	0.685 ± 0.074	0.695 ± 0.107	0.695	0.345 ± 0.135	0.379 ± 0.200	REVISIT
E009	Random + soil features	0.664 ± 0.049	0.643 ± 0.054	0.664	0.337 ± 0.083	0.312 ± 0.072	REVISIT
E010	TGB (major road)	0.711 ± 0.085	0.699 ± 0.081	0.711	0.384 ± 0.150	0.380 ± 0.130	REVISIT
E011	TGB tuned	0.725 ± 0.084	0.716 ± 0.081	0.725	0.447 ± 0.184	0.408 ± 0.147	REVISIT
E012	TGB tuned (expanded)	0.730 ± 0.085	0.724 ± 0.081	0.730	0.420 ± 0.170	0.413 ± 0.152	REVISIT
E013	Hybrid bias-corrected	0.768 ± 0.069	0.742 ± 0.070	0.768	0.507 ± 0.167	0.458 ± 0.126	SUCCESS

Table 3: Experiment progression E007–E013 (the ladder under examination).

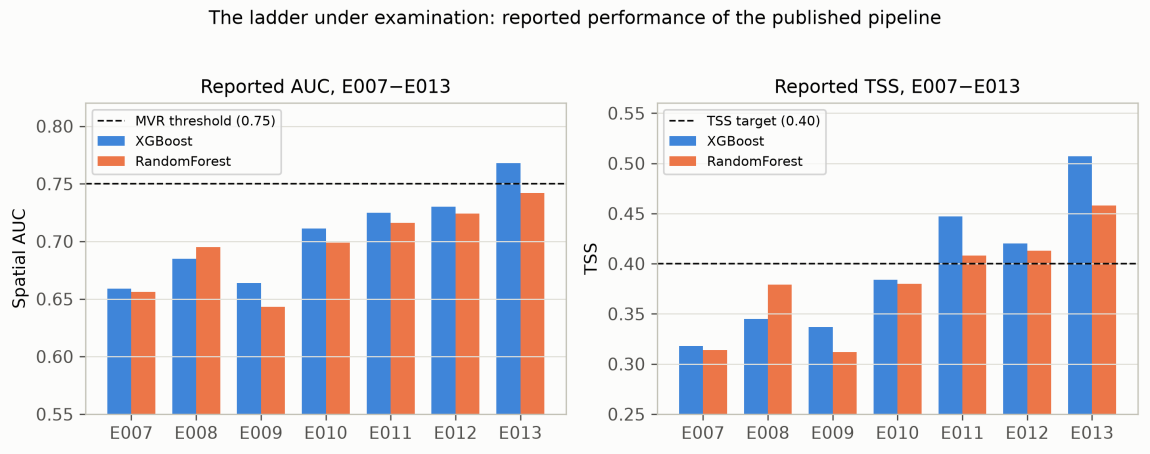


Figure 2: Reported AUC and TSS across the seven experiments of the published pipeline (E007–E013; Table 3). This is the ladder under examination; Section 3.2 asks whether the rise *between* E007 and E013 is real.

rithms, whereas adding the river feature contributes +0.0424 **AUC**, positive in 12/12 paired comparisons (the redesign contribution is +0.0054 on the terrain-only feature set and −0.0142 on the full set; we state the feature set so a reader does not encounter the difference unannounced). The hybrid design beats the random design in 3/3 algorithms when each is scored on the *hybrid* evaluation background, and in **0/3** on each of the uniform, target-group and stratified evaluation backgrounds. The advantage evaporates the moment it is judged anywhere but at home.

How large is judging at home? For the hybrid design — the top rung of the ladder — the inflation from own-background evaluation is positive in **60/60** seed×algorithm cells (mean +0.037, range +0.005 . . . + 0.084), the same order of magnitude as the entire reported E007→E013 gain. This home-court inflation is specific to that design, and we state the scope rather than let it read as general: the TGB design scored *higher* on the uniform background than on its own (mean −0.0054, 22/60 cells positive). The TGB rungs of the ladder (E010–E012) are therefore invalidated not by home-court inflation but

Model	AUC	Std	Gap vs E013
Random (chance)	0.500	0.000	+0.251
Heuristic (river < 2 km)	0.581	0.054	+0.170
DKNS (spatial interpolation)	0.646	0.032	+0.105
E013 XGB (seed-avg)	**0.751**	0.013	—
E013 XGB (best run)	0.768	0.069	—

Table 4: Null-model comparison under spatial CV. Gaps are computed against the E013 seed-average (0.751), not the best run, because a single run is not a reportable quantity (Section 3.7); the submitted version quoted these gaps against the best run (0.768), which overstated each by 0.017. The nulls are scored on a regular grid of all cells, while the E013 column is the model's own-background seed-average — so the margins shown still mix evaluation backgrounds, a defect corrected in Section 3.2, where the consequence for this level comparison is stated. DKNS (distance to known nearest site) uses site locations as an unfair upper bound.

by the common-background comparison itself, which contributed approximately zero once identified. Aggregated over the synthetic worlds, a model scored on its own background is reported higher than its truth in **343/360 runs (95.3%; median** +0.187**, minimum** −0.031**)** (Figure 3). We reject the published ladder gain (+0.092) directly: equivalence confidence intervals exclude it in **12/12** algorithm×evaluation-background cells, and a bootstrap that resamples presence *blocks* rather than points (30 replicates, fitted in-bag and scored out-of-bag) rejects it for all three algorithms, with CI upper bounds of +0.008, +0.025 and +0.026. Thirty replicates is a small bootstrap; we report it because the rejection margin is wide relative to the resampling noise, not because the interval endpoints are precisely estimated. Changing the MaxEnt regularisation β across 0.5–4.0 changes nothing (−0.020 to −0.022, 1/10 runs positive), so the artefact is not an overfit of an unregularised model.

One consequence for the level claim reported in Table 4 must be stated plainly. That table scored the E013 model on its own (hybrid) background but scored the null models on a regular grid of all cells; given the inflation above, the +0.105 seed-average margin over the DKNS null mixed the two backgrounds. Held to a common (uniform) background — the sampling on which the nulls were scored — the E013 design family reaches 0.706 (XGBoost, the primary learner, seed-mean; E218), which still exceeds the spatial-interpolation null (0.646), but the margin shrinks from +0.105 to roughly +0.06, and

the reported worst-seed margin (+0.083) moves toward the +0.03 detection floor of Section 3.6. The two figures are not measured under identical sampling — 0.751 comes from the published pipeline and 0.706 from the lattice-based benchmark (Section 2.6) — so the shrinkage should be read as an order of magnitude, not a difference of differences. The qualitative level claim — that the environmental features capture structure beyond nearest-site interpolation — survives the correction; its size does not. We therefore treat Table 4 as an upper bound on the model's advantage over the nulls, not a robust floor.

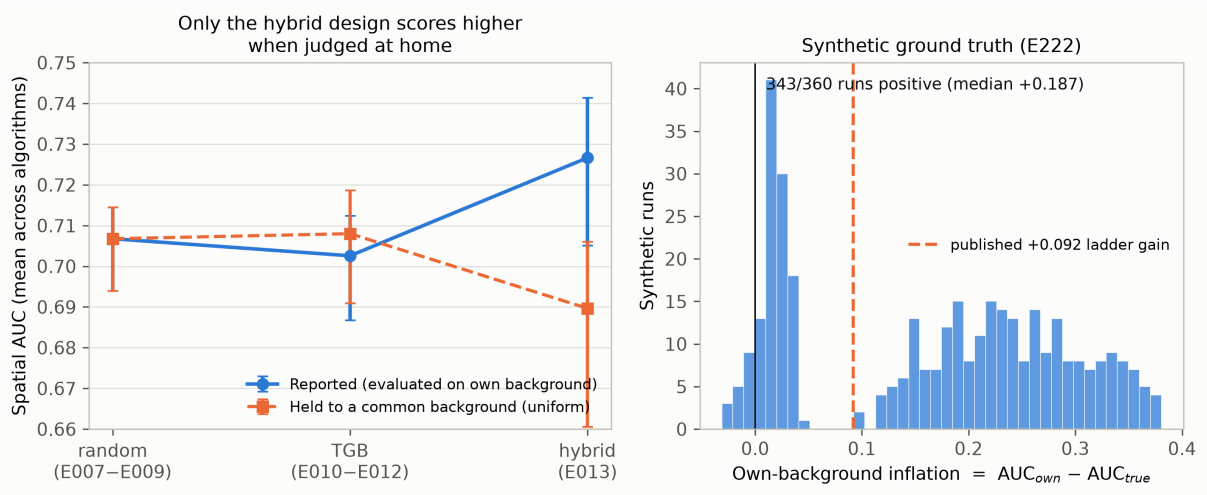


Figure 3: The evaluation artefact. Left: reported AUC (evaluated on a design's own background) against the same designs held to a common background (uniform). Only the hybrid design scores higher when judged at home. Right: own-background inflation across 360 synthetic runs of known ground truth (343/360 positive, median +0.187); the published +0.092 ladder gain is marked.

3.3 The selection criterion has no interior optimum

Why does the ladder exist at all? The selection criterion — AUC computed on a design's own background — has no interior optimum: its maximum always sits at the edge of whatever dial is swept, in both the synthetic and real sweeps (Figure 4). In the synthetic worlds the criterion rises monotonically along the hard-negative dial (0.7367 → 0.8904); on the real data it is monotone from 0.1 upward, with one dip of −0.0071 at 0.0 → 0.1. Because the criterion never points inward, it never tells a researcher to stop.

The practical consequence is mild at the operating point our manuscript used. We stopped at `hard_frac` = 0.30 because our grid stopped there, not because the criterion

said to; the cost of that stop, against the truth-best configuration on the same grid, is small (+0.0044 real; median +0.0000 synthetic, mean +0.0012, maximum +0.0088), and on the manuscript's own grid the criterion does not pick the truth-best design in 29/60 cases but the differences are essentially zero. The danger is structural: if the dial were extended, the cost is large (+0.0973 real; +0.1937 synthetic median, 100% positive), and the criterion provides no brake. We note what the criterion does *not* do, correcting an earlier internal claim: the design it selects (hybrid(1.0)) is never the truth-worst design (0/60 cases); the truth-worst is hybrid(0.0) in 50/60, hybrid(0.3) in 8 and hybrid(0.7) in 2 of 60 cases. The selected design costs +0.194 **$\mathbf{AUC_{true}}$ against the best available**, but it is not the worst.

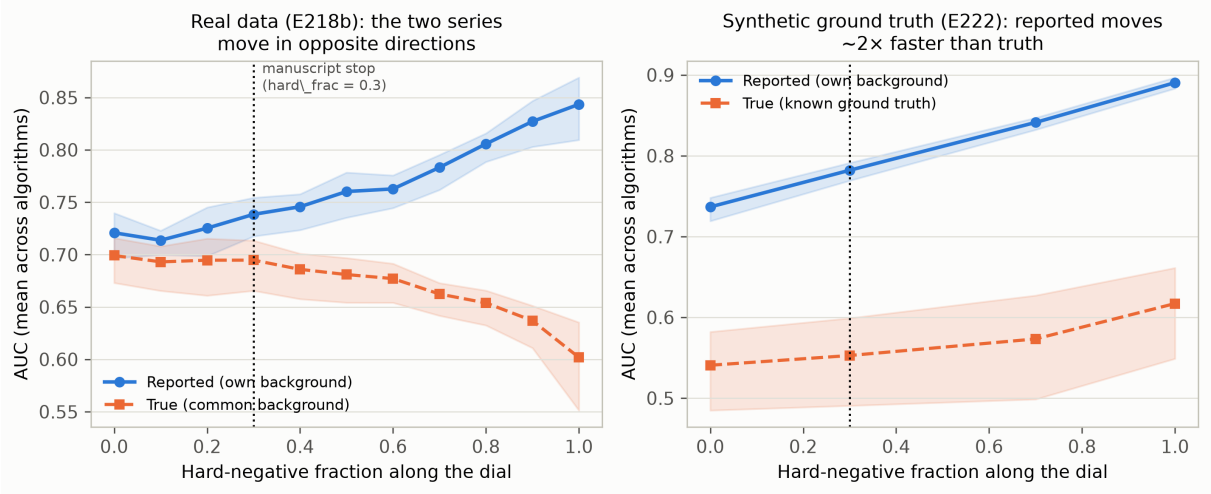


Figure 4: The selection criterion on a design's own background (solid) keeps rising to the edge of the dial; the true performance (dashed) does not, so the criterion has no interior optimum. Left: real data — the two series move in opposite directions. Right: synthetic ground truth — reported performance moves about twice as fast as truth. The dotted vertical line marks the manuscript's stop at `hard_frac` = 0.30; the dial ends at 1.0.

3.4 Reported and true performance diverge

In the synthetic worlds the reported number moves about **twice** as fast as the truth along the dial (+0.1538 reported vs +0.0764 truth; endpoint ratio 2.01×, per-run OLS slope 2.12×, per-run median ratio 2.00× — we name the estimator because the three definitions bracket "about twice"). On the real data the two move in **opposite directions** (+0.1227 reported vs −0.0973 common-background), which is a stronger statement and needs no

magnification. A pre-registered prediction that own-background inflation rises with the hard-negative fraction in every regime **failed** (pooled Spearman 0.4395 $<$ 0.5) and is reported as failed.

3.5 The target-group-background null, reported as unexplained

Our synthetic worlds gave TGB exactly the condition its theory requires — a known, survey-biased observation process — and TGB did not help. Over the synthetic runs, TGB minus random on top-decile map agreement (Jaccard) is -0.010, positive in 46.7% of paired comparisons. In regional-bias regimes TGB is neutral rather than harmful (World C -0.0010, 56.7% positive; World D $+0.0022$, 73.3% positive), and it wins only under model misspecification (Jaccard 0.4504 vs 0.4458 in World B), which we report because it is true but do not build on.

We proposed an explanation for this uncomfortable null and nearly wrote it into the manuscript: because `road_dist` is not a model feature, the survey-bias factor is unrepresentable in feature space, so TGB has nothing to cancel — a tidy, "predicted" null. We pre-registered a test with both decision branches written first, then ran it (E224): the same synthetic worlds with `road_dist` added to the features. It made no difference. TGB minus random was -0.0217 with `road_dist` in the features (30% positive) and -0.0254 without (30% positive). We therefore report the null as **unexplained**, and we disclose the limitation of our own test: `road_dist` correlates $+0.49$ with `river_dist` (and $+0.31$ with elevation), so the model already had partial access to the bias signal in the control condition; a clean test needs a survey-effort surface orthogonal to terrain by construction, which is future work. We resisted re-interpreting the secondary metric (`auc_true`, which did rise from 36.7% to 60% positive) because the pre-registered primary metric was map Jaccard, and map Jaccard said no.

3.6 Confidence intervals and a detection floor

The equivalence CIs that reject the $+0.092$ gain (Section 3.2) also bound what we can rule out: at $n = 378$ the spatially blocked bootstrap cannot exclude effects below about

+0.03. Findings smaller than that are indeterminate at this sample size, not absent.

3.7 Priority maps are unstable under seed, and the protocol that fixes them

A model whose numbers are artefactual still produces a priority map, and in heritage practice the map is what gets used. Across random seeds, 28.1%–47.4% of top-decile cells turn over between seed pairs — a single-seed map is not a defensible basis for a survey decision. The remedy is cheap and we now make it a protocol: seed-ensemble. The number of seeds required to stabilise the map is $k^* = 2$–5 for Jaccard ≥ 0.85, 4–7 **for** ≥ 0.90, and 7–9 for ≥ 0.95; we recommend $k \geq 7$ (Figure 6). This ensemble reveals a useful distinction between a *robust core* (the consistent core area) and a *contingent fringe* (the peripheral area that depends on the background design). The robust core is defined as the cells that fall in the top 10% under all three background designs (random, TGB and hybrid), each map being the average of 10 seeds; the contingent fringe is the cells that fall in the top 10% under exactly one design. Observed site density is higher in the core than in the fringe: 1.93× for RandomForest (30.7 vs 15.9 sites per 1,000 km^2), 4.34× for XGBoost (40.8 vs 9.4) and 5.62× for MaxEnt (31.7 vs 5.65; Figure 5). This difference indicates spatial consistency, not independent validation, because the sites used to compute these densities were also used to train the models.

3.8 Corrections to our own diagnostics: the volcano inventory and a correlation that does not reproduce

Two corrections to our own diagnostics, none prompted by the reviewers.

The volcano inventory. The analysis code hard-coded 7 volcanic centres; the canonical inventory contains **13** inside the paper's own 111–115°E bounds (adding Lawu, Wilis, Kawi-Butak, Penanggungan, Iyang-Argapura, Baluran), and two of the additions sit inside the Malang–Mojokerto site concentration. `volcano_dist` is not a training feature, so model AUCs are unaffected; the Test 1 tautology diagnostic is. Recomputed on the

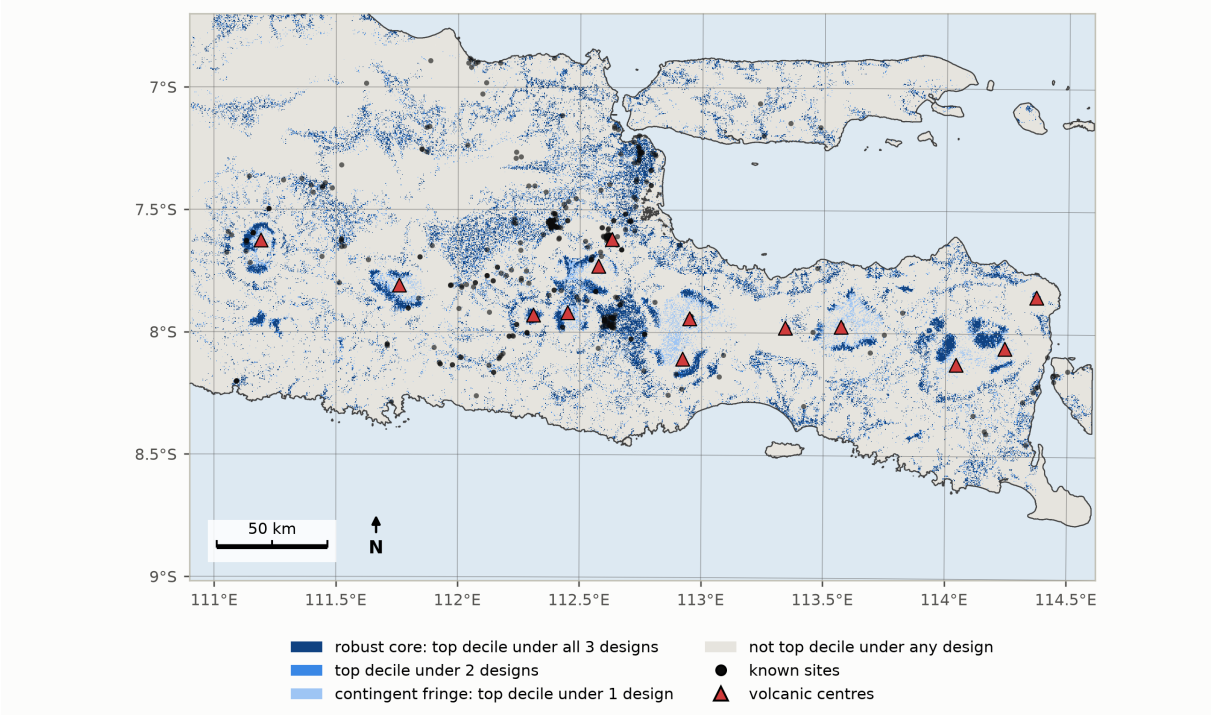


Figure 5: Robust core and contingent fringe for XGBoost under a 10-seed ensemble, cells coloured by how many of the three background designs placed them in the top decile. The core concentrates on the accessible lowlands that hold the known record; the contingent fringe is the remote upland on which background assumptions disagree (median 1.0 km from roads and 1,107 m elevation in the MaxEnt fringe).

canonical inventory, the suitability–volcano-distance Spearman correlation is $\rho = -0.281$ (legacy 7-volcano re-run: -0.243). The verdict is unchanged: $|\rho| < 0.5$, so the test still passes.

A published number that does not reproduce. The submitted manuscript reported $\rho = -0.163$ for that same correlation, taken from a single model instance. A five-seed re-run on the same 7-volcano inventory gives -0.243. This is the seed instability of Section 3.7 appearing *inside* the manuscript's own tautology diagnostic, and it is direct internal evidence that the seed-ensemble protocol is needed, not optional. The corrected values reported here are seed-ensembled; the E007–E013 ladder in Table 3 is the original published pipeline, reproduced as the object under examination.

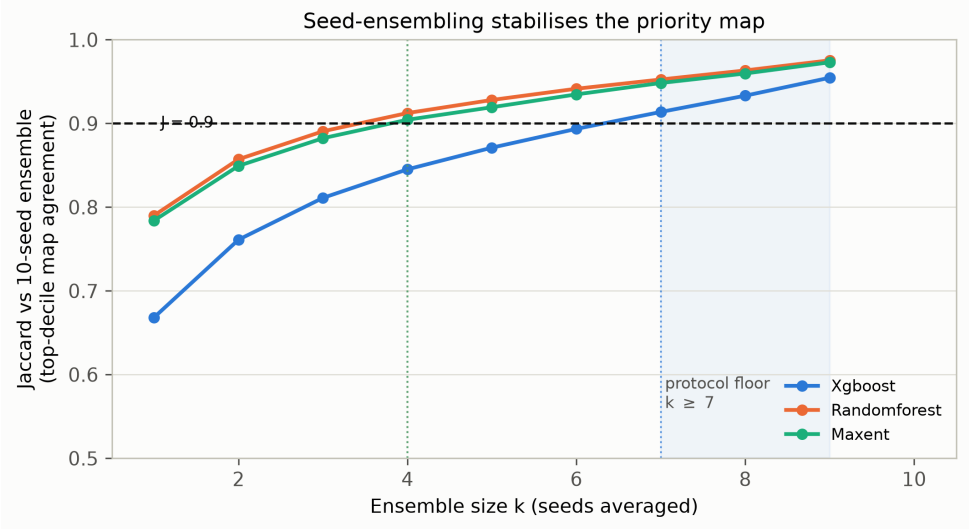


Figure 6: Agreement of an ensemble of k seeds with the full 10-seed ensemble on the top-decile priority map, one line per learner. A single seed agrees at Jaccard $\approx$ 0.65–0.80; $k^* = 4$–7 suffices for Jaccard ≥ 0.90, and $k \geq 7$ is the protocol floor (shaded).

3.9 Volcanic setting or elevation?

This analysis tests whether the observed suitability pattern reflects a volcanic setting or simply the effect of elevation. Distance to a volcanic centre (`volcano_dist`) is not used as a training feature, so any pattern in the model's predictions that relates to a volcanic setting must come through the terrain variables the model uses. Near land is defined as land within 20 km of a Quaternary volcanic centre, and far land as land 40 km or more from one; the analysis includes the 30 canonical volcanic centres, including those outside the map frame. The comparison is made within each 200 m elevation band, using coarsened exact matching on slope, the terrain ruggedness index (TRI) and the topographic wetness index (TWI). The analysis uses the stored 10-seed ensemble maps, without retraining any model. Uncertainty is estimated with a spatial block bootstrap of 1,000 replicates; the decision threshold of 0.05 was set before the analysis, and the design was pre-registered in the public repository. The verdict labels follow the pre-registered rule: SPECIFIC if the 95% CI excludes zero and the size of the difference is at least 0.05; NOT SPECIFIC if the whole 95% CI lies between -0.05 and $+0.05$; and INCONCLUSIVE for any result that meets neither criterion.

Above 1,000 m the study area contains about 3,435 km^2 of land, of which 97.5% lies

within 20 km of a Quaternary volcanic centre and only 0.4% lies 40 km or more from one (Figure 7A). Land far from Quaternary volcanic centres covers only about 15 km^2 at 1,000–1,200 m and none above 1,200 m. This means that, in this study area, the effects of elevation and of proximity to a Quaternary volcanic centre cannot be adequately separated above about 1,000 m. This study therefore does not attribute the pattern there specifically to either elevation or volcanic setting. This limit of separability can shift between 800 and 1,200 m, depending on the minimum-area criterion used.

Below 1,000 m, the difference in suitability between land near and far from Quaternary volcanic centres is +0.049 for the random design (95% CI +0.024 to +0.077), +0.049 for TGB and +0.060 for hybrid (95% CI +0.030 to +0.091; Figure 7B). Under the pre-registered rule, the hybrid result meets the criterion for SPECIFIC, but the summary across designs remains INCONCLUSIVE, because two of the three designs do not give a firm conclusion under that criterion. Additional post hoc analyses show that this difference ranges from +0.01 to +0.07, depending on the zone definitions and the matching procedure; when distance to rivers is also matched, the difference falls to +0.028. The far comparison land lies mostly in the Southern Mountains, an old volcanic arc of Eocene–Miocene age with karst landforms (Smyth, Hall & Nichols 2008), so it would not be correct to call it non-volcanic. In addition, the MaxEnt contingent fringe in Figure 5 has a median elevation of 1,107 m, above the limit of reliable separation; that pattern therefore cannot be interpreted specifically as either an elevation effect or a volcanic effect. The limited power of the test also does not allow a firm conclusion about whether the disagreement between designs differs between near and far land.

3.10 The taphonomic claim is withdrawn, and a quota design that failed

The taphonomic interpretation we built on the ladder — that low predicted suitability near volcanoes is evidence of buried settlement — is withdrawn, because the model comparison that supported it does not survive the common-background benchmark (Section 3.2). Low predicted suitability near a volcano is not evidence of a buried site, in

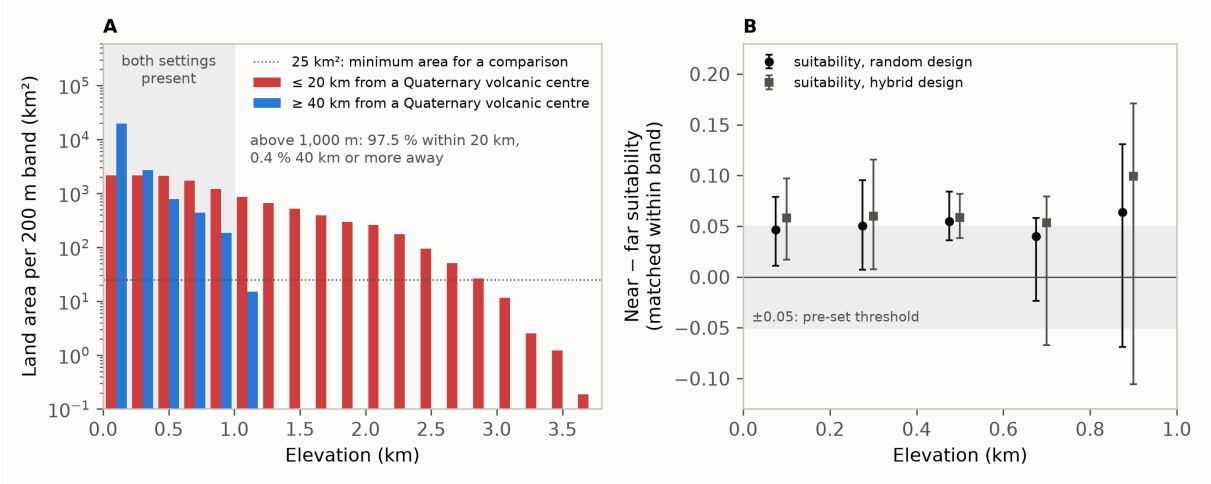


Figure 7: (A) Land area in each 200 m elevation band for land near ($\leq$ 20 km) and far ($\geq$ 40 km) from a Quaternary volcanic centre. (B) Difference in suitability between near and far land after matching on elevation, slope, TRI and TWI, with 95% confidence intervals (XGBoost); the grey band marks the decision threshold of ± 0.05.

the reviewer's own terms (Section 1.2: mechanism 3 cannot be read off a suitability surface). We also report, without softening, a quota background design we tested in its home regime (regional survey bias) and in its most favourable regime (balanced-regional truth): it failed in both (World C: -0.2457 AUC_{true}, -0.4688 Jaccard, 0/30; World D: -0.2027 AUC_{true}, -0.2826 Jaccard, 0/30).

4 Discussion

4.1 The lesson: hold the evaluation background fixed

The central methodological finding is that a background-design comparison is not identified unless the evaluation background is held fixed across the designs being compared, and that the artefact this produces is large enough to masquerade as a result. We are not the first to observe that background choice drives reported performance (Hijmans 2012; Lobo, Jiménez-Valverde & Real 2008; Phillips et al. 2009); what we contribute is the mechanism (a selection criterion with no interior optimum, Section 3.3) and a direct quantification of the inflation it produces (95.3% of synthetic runs; for the hybrid design, 60/60 real seed×algorithm cells). The practical recommendation follows directly: *report background designs on a common evaluation background, and state the evaluation availability domain*

of every metric. This is the accessible-area requirement of Barve et al. (2011) applied to evaluation rather than to fitting: the domain over which a metric is computed is part of the metric, and a number reported without it cannot be compared with another. That is the lesson of this line of work, and we learned it about our own model.

4.2 The archaeological content is in the background

What makes a presence–background model archaeological, rather than a generic classifier run on coordinates, is not the label set. It is the background design: a statement about where a survey could plausibly have detected a site had one been present. That is what TGB and hybrid designs encode, and it is why they are archaeological choices rather than sampling conveniences. The sharp corollary, which this paper exists to state, is that because the archaeological content lives in the background, and because the model was evaluated against that same background, the very thing that made the design archaeological is what contaminated its evaluation.

4.3 A suitability surface is not a discovery map

With the site-prediction claim withdrawn, the output is a suitability surface with declared limits, not a map of where sites are. Low suitability does not mean "buried" or "destroyed"; high suitability does not mean "a site is there" (Section 1.2). This restraint is now stated in the manuscript in these words, and it is a constraint the revised paper takes seriously rather than a caveat appended to an overclaim.

4.4 Implications for heritage management in East Java

Under Indonesia's Cagar Budaya framework (UU 11/2010), BPCB Jawa Timur surveys inform permitting and rescue prioritisation. A suitability surface could in principle support that workflow, but only under two conditions this revision makes explicit. First, because priority maps turn over by 28–47% between seeds (Section 3.7), a single-seed map is not a defensible basis for a permit; the seed-ensemble protocol ($k \geq 7$) is a precondition for operational use. Second, because the model reports suitability and not

presence, the robust core (1.9–5.6× the site density of the fringe) is a prioritisation aid, not a prediction: it ranks where to look, not what will be found.

4.5 The corrected protocol

We summarise the protocol that follows from the above, which is the constructive contribution of the revision:

1. **Hold the evaluation background fixed** when comparing background designs; report own-background numbers only alongside common-background numbers.

2. **Declare the availability domain** of every metric (background design, benchmark, selection rule).

3. **Seed-ensemble priority maps** ($k \geq 7$); report a robust core and contingent fringe, never a single-seed map.

4. **Treat the selection criterion as unbraked**: because it has no interior optimum, justify any stop on the design dial externally, not by the criterion's own maximum.

4.6 Limitations

1. **Calibration gap between synthetic and real.** At the manuscript's operating point the selection criterion picks hybrid on real data but random on synthetic data, so the synthetic worlds do not reproduce the real-data selection behaviour there. This limits transferring the E222 conclusions back to the real case, and is stated here rather than hidden in a supplement.

2. **The truth-slope sign is regime-contingent** — negative on real data, positive on synthetic. We do not claim "hard negatives are always bad."

3. **Four synthetic regimes are not a universal result.** Across the four worlds (road-bias, misspecified, regional-bias, balanced-regional-truth; $n \approx 300$–500), no background design exceeded uniform on truth recovery by a meaningful margin

(aggregate difference ≈ -0.0004 AUC; TGB scored above random in 27/60 runs). This is not a refutation of target-group background in general (Phillips et al. 2009); it is a result for the regimes and sample sizes tested.

4. **A detection floor of** $\sim +0.03$ at $n = 378$: smaller effects are indeterminate, not absent (Section 3.6).

5. **The Boyce index is a direction check, not a selector.** Its optimum is uncalibrated to truth (median $+0.50/+0.54$, borderline); we use it only to confirm direction, and we have aligned the audit flag with this narrative so the two do not contradict.

6. Above about 1,000 m (the limit shifts between 800 and 1,200 m, depending on the minimum area used), **elevation and proximity to a Quaternary volcanic centre cannot be adequately separated** in the study area; the model's prediction pattern there, including the high-elevation fringe in Figure 5, therefore cannot be attributed specifically to either factor (Section 3.9).

7. **The two-stage "suitable but absent" decomposition** is declared future work: it tests *why* sites are absent, which requires the site-prediction claim we have withdrawn, and burial-depth data at survey resolution are not yet available.

8. **External transfer beyond East Java is untested.**

5 Conclusions

The AUC ladder we reported — from 0.659 (E007) to 0.751/0.768 (E013), attributed to increasingly realistic pseudo-absence design — is an evaluation artefact. Each background design was scored against the negatives it had selected for itself; judged on a common evaluation background, redesigning the background contributes -0.014 AUC and adding one hydrological feature contributes $+0.042$. The selection criterion that produced the ladder has no interior optimum, so it never tells a researcher to stop, and reported and true performance move in opposite directions on the real data. We withdraw the taphonomic

interpretation we built on the ladder, and we correct two defects in our own diagnostics (the volcano inventory and a correlation that does not reproduce across seeds).

The contribution that remains is the diagnosis and the protocol: compare background designs only on a held-fixed evaluation background, declare the availability domain of every metric, and seed-ensemble priority maps ($k \geq 7$). We offer these not as a consolation for a lost headline but as the more useful finding the reviewer's request produced: a mechanical account of why presence–background comparisons can look decisive when they are not, and a cheap, checkable protocol that keeps the next comparison honest. We have applied it to our own work, including to the internal revision drafts that preceded this manuscript, where the same failure mode recurred and only a blind re-derivation caught it.

Supplementary Materials

Supplied as a single document, `supplementary_tables_v0.3.pdf`. Every table in it is generated directly from the raw result files by `build_supplement.py` and `build_e235_tables.py` (included in the code repository); no value is retyped.

- **Table S1:** Seed stability — XGBoost and RandomForest AUC and TSS across the 20 alternate seeds for E013.
- **Table S2:** Block-size sensitivity — spatial CV at ∼40, ∼50 and ∼60 km, with 95% intervals.
- **Table S3:** E217 common-background benchmark — every design × feature set × algorithm, scored on its own background and on a common background.
- **Table S4:** E218 own- versus common-background matrix — three training designs against four evaluation backgrounds.
- **Table S5:** E222 synthetic worlds — reported against true AUC by background configuration and algorithm.

- **Table S6:** Per-experiment design matrix for E217–E224 (extends Table 2), including which experiments used `road_dist` as a feature.
- **Table S7:** E235 elevation bands — land near and far from Quaternary volcanic centres in each 200 m band, and the matched near − far differences (Section 3.9).
- **Table S8:** E235 pooled differences — by zone definition and algorithm, with the pre-registered verdicts.

Acknowledgements

The authors thank the editor and two reviewers; the request to benchmark against Maximum Entropy produced the finding this paper reports.

Funding

This research received no external funding.

Competing Interests

The authors declare no competing interests.

Authors' Contributions

M.A. conceived the research design, developed the methodology, implemented software, conducted formal analysis, curated data, created visualisations, and wrote the original draft. G.F.G. contributed to methodology refinement, code review, and manuscript editing. Both authors approved the withdrawal of the central claim reported in this revision.

Data Availability

Archaeological site data: the compiled East Java site dataset is in the repository at `data/processed/east_java_sites.geojson` (378 with valid covariates in bounds).

Sources: OpenStreetMap (ODbL), Wikidata (CC BY-SA), Wikipedia Indonesia, BPCB Jawa Timur publications.

Environmental covariates: Copernicus GLO-30 DEM and derived layers from the Copernicus Programme (ESA), `https://dataspace.copernicus.eu/`. OpenStreetMap road and river data, `https://openstreetmap.org` (ODbL).

Code: all source code, including E217–E224 and the verification script, is at `https://github.com/neimasilk/volcarch-repo` (MIT licence).

The G1 verification script is `verify_headline_numbers.py`.

Python dependencies are pinned in `requirements_submission_lock.txt`.

Interactive dashboard: `https://volcarch-dashboard.streamlit.app`.

AI Disclosure

This research used an AI-augmented workflow (Claude, Anthropic) throughout, and we describe its extent rather than minimise it, because a paper about not claiming more than the evidence supports should not begin by understating its own method. The AI assistant was used for literature synthesis, statistical and geospatial scripting, figure generation, drafting and editing of this manuscript, and — materially — for drafting the pre-registered designs of the refutation experiments reported in Section 2.6, which are archived with authorship recorded in the repository cited under Data Availability. It was also used as an adversarial reader: one review pass was run with the assistant instructed to refute the manuscript's claims from the raw output files. The research questions, the decision to withdraw the central claim, the domain interpretation, and every final scholarly judgement are the authors'. The revision's disclosures arose from the same workflow rather than despite it: the corrections listed in the Response to Reviewers were found by a mechanical, blind re-derivation of every headline number from the raw per-run outputs, run as a pre-submission gate (script cited under Data Availability), or from the stored experiment results where no per-run file exists. All results, code and text were reviewed and verified by the authors, who take full responsibility for the content.

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